

VAPO: Value-Model-Based RL for Advanced Reasoning — 60.4 on AIME 2024

Headline Result

60.4 on AIME 2024 (avg@32)

Prior SOTA Baselines

DAPO: 50 | DeepSeek-R1-Zero-Qwen-32B: 47

Base Model & Data

Qwen2.5-32B, no SFT data

Efficiency & Stability

~5,000 steps, zero training crashes

Source: ByteDance Seed, "VAPO" (arXiv:2504.05118)

Why Value Models? Finer Credit Assignment for Long Chain-of-Thought Reasoning

Value-model-free (GRPO/DAPO)

- Trajectory-level reward; advantage assigned uniformly across tokens
- Simple and effective baseline for many tasks
- Weak token-level credit assignment in long reasoning chains
- Higher variance from Monte Carlo-style return estimates

Value-model-based (VAPO)

- Token-level value estimation for precise credit assignment
- Traces each action's impact on future returns
- Lower-variance per-token estimates improve training signal quality
- Better sample efficiency when value model generalizes across samples

Three Challenges Plaguing Value-Model-Based RL in Long-CoT Tasks

1) Value Model Bias

Long trajectories + bootstrapped learning amplify value estimation error

Why it matters

Can destabilize policy learning and degrade final reasoning accuracy

2) Heterogeneous Sequence Lengths

Short and long responses need different bias-variance tradeoffs

Why it matters

Fixed GAE setting is suboptimal across diverse response lengths

3) Reward Signal Sparsity

Only final <eos> token receives reward in this setup.

Why it matters

Sparse delayed supervision weakens token-level learning signals.

VAPO Framework: Integrated Solutions for Value-Model-Based Reasoning RL

Challenge 1:
Bias

Value-Pretraining

Mitigates initial value bias

Decoupled-GAE

More robust value estimation

Challenge 2:
Length Heterogeneity

Length-adaptive GAE

Dynamic λ by response length

Challenge 3:
Reward Sparsity

Group-Sampling

Multiple samples for better baseline

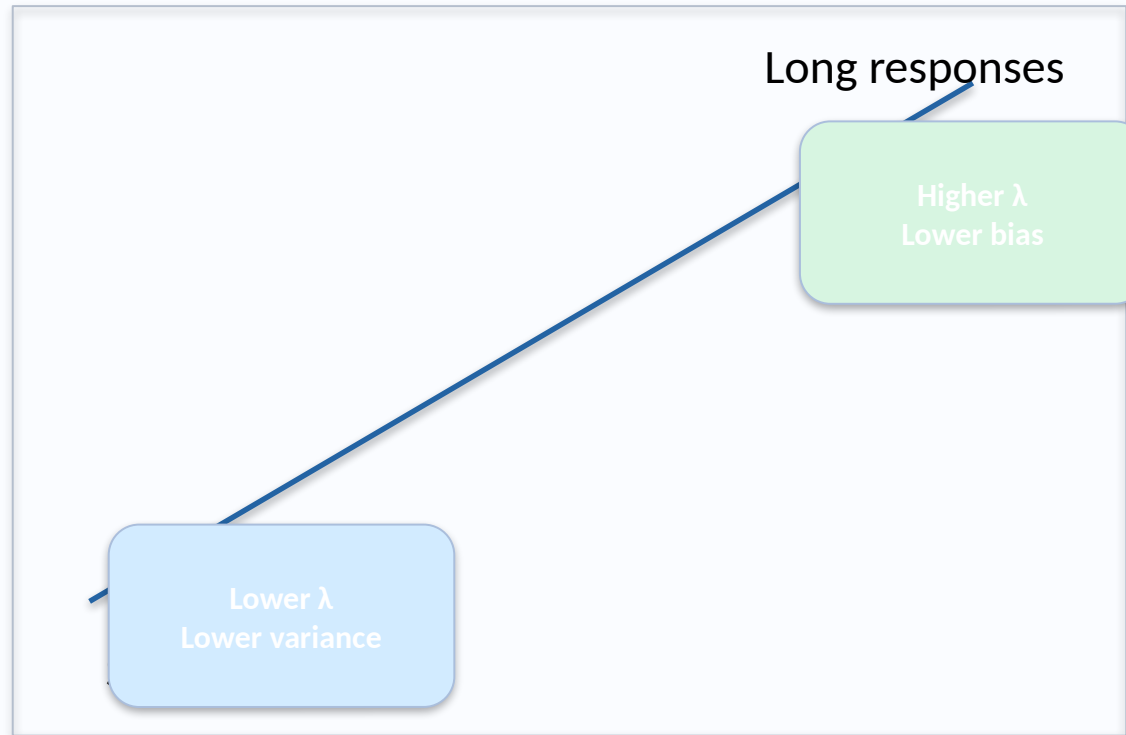
Token-level Loss

Token-level normalization

Positive Example LM Loss

Self-imitation on correct trajectories

Length-Adaptive GAE: Dynamic λ Adjustment for Variable Response Lengths



Core idea: λ is adjusted as a function of response length.

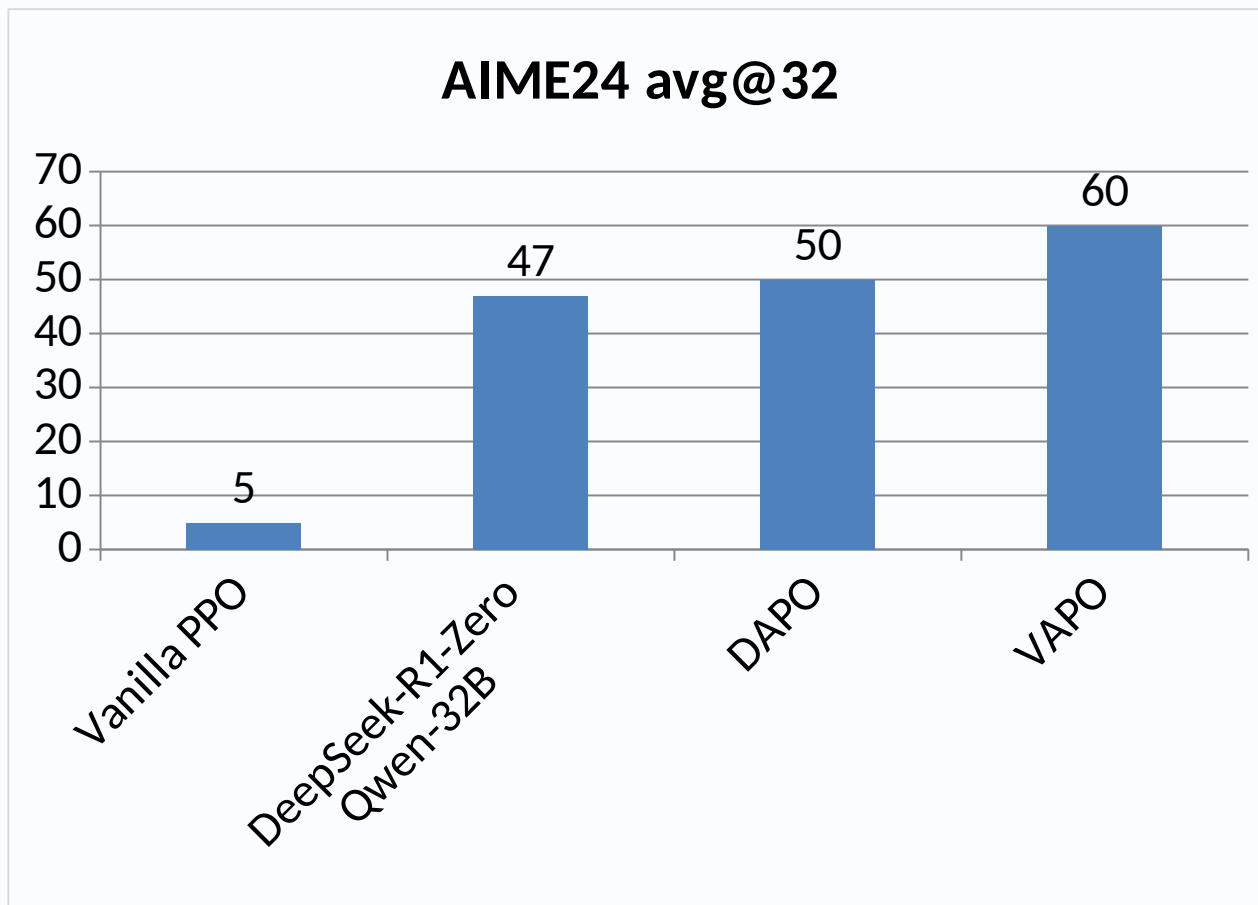
Short responses use lower λ to reduce variance in advantage estimates.

Long responses use higher λ to reduce bias over long-horizon dependencies.

Addresses heterogeneity challenge unique to long-CoT reasoning.

Ablation impact: removing Length-adaptive GAE lowers AIME24 from 60 to 45.

VAPO Scores 60 on AIME 2024 — 10+ Points Above DAPO and DeepSeek-R1-Zero



Main result (AIME24 avg@32):

- Vanilla PPO: 5
- DeepSeek-R1-Zero-Qwen-32B: 47
- DAPO: 50
- VAPO: 60 (reported headline: 60.4)

VAPO exceeds DAPO by 10 points and DeepSeek-R1-Zero by 13 points.

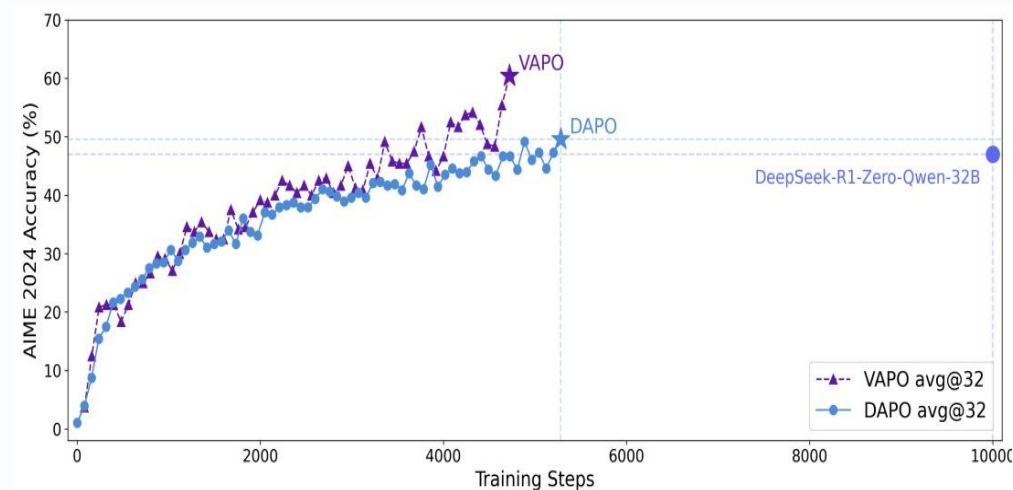
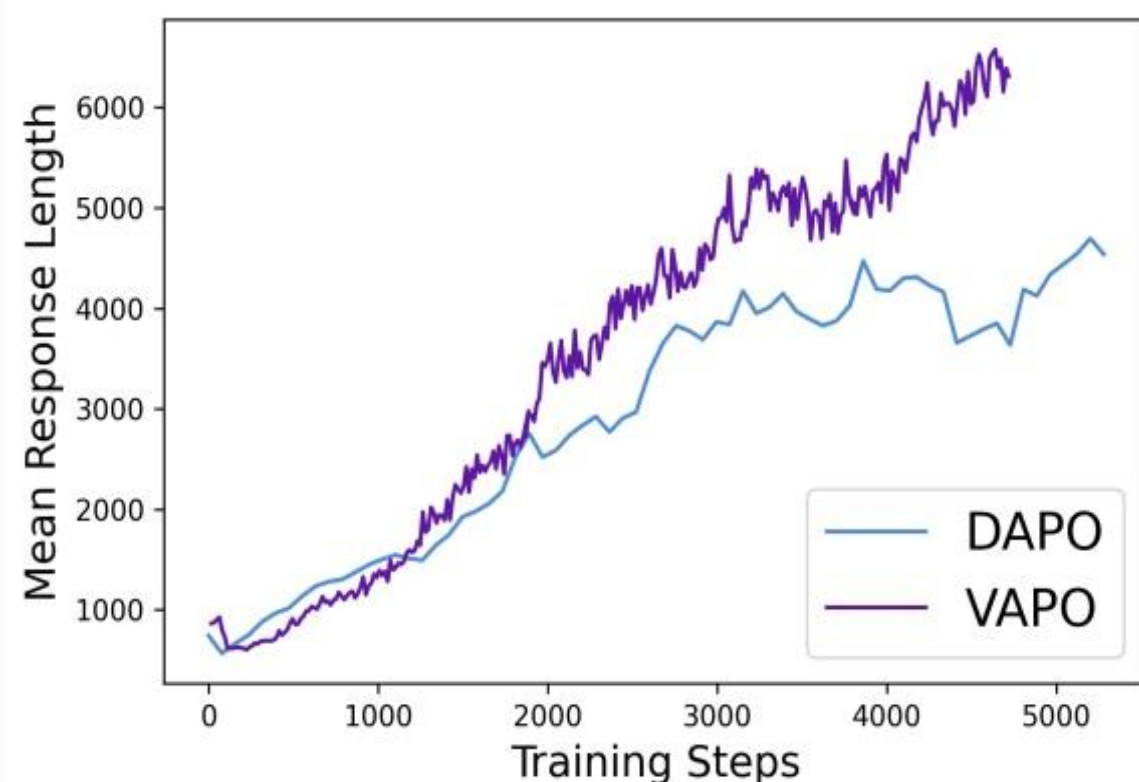
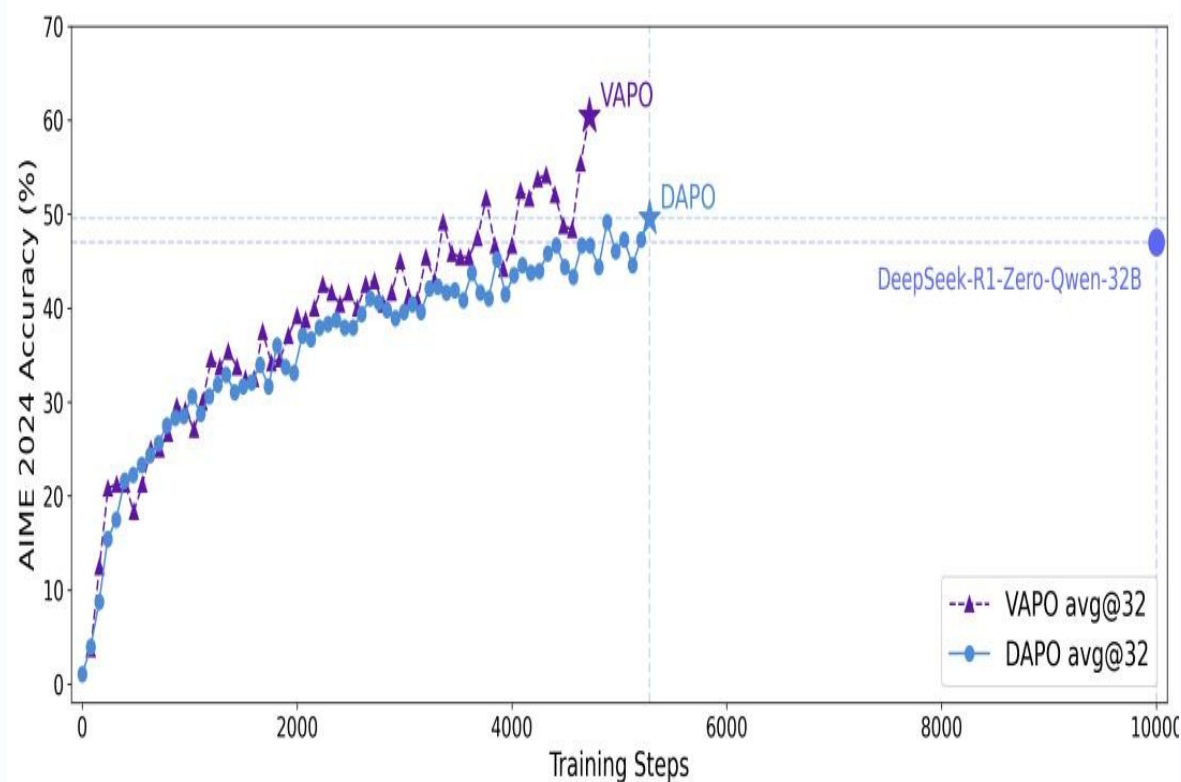


Figure 2: Training curve (VAPO vs DAPO)

VAPO Reaches SOTA in 5K Steps — Half the Training Budget of DAPO



VAPO reaches ~60% around step 4,500; DAPO plateaus near 50% around step 5,500.
DeepSeek-R1-Zero-Qwen-32B reaches 47% after approximately 10,000 steps.
VAPO responses grow to ~6,000 tokens vs ~4,500 for DAPO, indicating longer reasoning chains.

Ablation: Every Component Matters — Value-Pretraining Alone Adds 49 Points

Configuration	AIME24 avg@32	Δ vs Full
VAPO (full)	60	0
w/o Value-Pretraining	11	-49
w/o Decoupled-GAE	33	-27
w/o Length-adaptive GAE	45	-15
w/o Clip-Higher	46	-14
w/o Token-level Loss	53	-7
w/o Positive Example LM Loss	54	-6
w/o Group-Sampling	55	-5

Largest impacts:

- 1) Remove Value-Pretraining: 60 $\rightarrow$ 11 (-49)
 - 2) Remove Decoupled-GAE: 60 $\rightarrow$ 33 (-27)
 - 3) Remove Length-adaptive GAE: 60 $\rightarrow$ 45 (-15)
- Interpretation: value pretraining is foundational; adaptive credit estimation compounds gains.

Key Takeaways: Value-Model-Based RL Is Back — With the Right Engineering

VAPO achieves 60.4 on AIME 2024, surpassing DAPO (50) and DeepSeek-R1-Zero-Qwen-32B (47) by 10+ points.
Value-model-based RL can outperform value-model-free methods when bias, length heterogeneity, and reward sparsity are jointly addressed.
Value-Pretraining is the most critical ingredient (60 → 11 without it).
Length-adaptive GAE effectively handles variable response lengths and contributes +15 points (45 → 60).
VAPO is training-efficient and stable: ~5,000 steps to SOTA with zero crashes across multiple runs.
Implication: future reasoning RL stacks should treat value modeling as a first-class, engineered subsystem.